AOC-0 Interim orbital capsule

Specifications and assembly instructions

Description

The AOC-0 is a low tech interim capsule that can be used for short suborbital or orbital missions. The capsule is battery powered and has a redundant short range radio transceiver, limiting its orbital endurance.

Specifications

Parameter	Value	Unit	Notes
Mass	1210	Kg	
Total EC	310	Units	
EC endurance	134	Minutes	Hibernating
	64	Minutes	Full operation
EC consumption	0.0884	EC/sec	Full operation
	0.04	EC/sec	Hibernating

Assembly and maintenance instructions

Parts list

1x	Mk1 Command Pod

1x	Probodobodyne Stayputnik

1x	Heat Shield (1.25m)

2x	Mk2 Radial-Mount Parachute

2x	Reflectron DP-10

1x	2HOT Thermometer

1x	PressMat Barometer

Assembly procedure

- Start with Mk1 pod
  - Drain all monopropellant
  - Remove all equipment from inventory slots
  - Set spare parts to 5
- Install heat shield axially on bottom of capsule
  - Set ablator to 0
- Install Stayputnik computer axially on top of capsule
  - Stayputnik name tag on window side
  - Set "Hibernate on warp" to "Auto"
  - Set "Hibernate" to "Off"

    - Set battery flow priority to -1
- Install radio antennas radially on pod
    - Mirror symmetry
    - Aligned with flag's sides
    - Antennas should run full length of capsule
    - Touching stayputnik on top and heat shield on bottom
- Install parachutes radially on sides of pod
    - 2x symmetry with snap
    - On white lines of Mk1 in default skin
    - On top of pod, sides should touch end of pod, no part should be visible protruding on top of capsule
    - Angle 10
    - Altitude 750
    - Pressure 0.6
- Install barometer and thermometer below parachutes
- Barometer to left of hatch, thermometer to right
- Bottom of pod, bottom of instrument should touch heat shield

Post flight maintenance

Part	System	Action	Notes
Stayputnik	Battery	Fill	
	Command	Check hibernation	Hibernate to off Hibernate in warp to on
Mk1 pod	Battery	Fill	
	Monoprop	Empty	
	Spare parts	5	
	Food	Fill	
	Water	Fill	
	Oxygen	Fill	
	Carbon Dioxide	Empty	
	Waste	Empty	
	Waste water	Empty	
	Reaction wheels	Active/Normal	
Parachutes	Pressure	0.6	
	Altitude	750	
	Deploy	When safe	
Radios	Status	Enabled	

Pre mission checks

- Check parachute parameters
  - Pressure 0.6 Altitude 750
- Check radios
  - Both ON
- Check supplies
  - EC full
  - Life support > 20%
- Check computer
  - Not hibernating
  - Auto hibernate in warp

Systems

Electrical

Description

The electrical system is supplied by the Mk1 pod batteries. There is a secondary battery in the Stayputnik computer that is set to be drained after the main battery is drained. This is only a last resort reserve and re-entry should be initiated as soon as possible when running on this battery.

System consumption

Class	System	Consumption	Unit	Notes
Computer	**Total**	**0.0284**	**EC/sec**	
	Stayputnik	0.0284	EC/sec	Full operation, divide by 1000 for hibernation
Radios	**Total**	**0.02**	**EC/sec**	
	DP-10 (right)	0.01	EC/sec	
	DP-10 (left)	0.01	EC/sec	
Life support	**Total**	**0.04**	**EC/sec**	
Attitude	**Total**	**0.2**	**EC/sec**	
	Mk1 wheel	0.2	EC/sec	
Sensors	**Total**	**0.0134**	**EC/sec**	
	Thermometer	0.0067	EC/sec	
	Barometer	0.0067	EC/sec	

Total EC consumption

Situation	Consumption	Unit	Notes
Maneuvering	0.2884	EC/sec	Sensors inactive
Full op + sensors	0.102	EC/sec	
Full operation	0.0884	EC/sec	
Hibernating	0.06	EC/sec	

Situation	Consumption	Unit	Notes
Hibernating, 1 radio	0.05	EC/sec	
Hibernating, no comms	0.04	EC/sec	Requires crew to get out of this state

Life support

The life support is a standard TAC LS unit taking supplies from the Mk1 capsule. This is enough to keep a kerbonaut alive for 3 days. This is much greater than the availability of electrical power (even in deep hibernation). The availability of power is the limiting factor and the capsule's endurance is 2 hours.

Resource	Endurance	Unit	Notes
Oxygen	18	hours	
Water	18	hours	
Food	18	hours	
Electric power	2.15	hours	Limiting factor, full hibernation

Attitude control

Reaction Wheels

The only way of controlling attitude is in the Mk1 reaction wheel. Loss of the wheel prevents the capsule from exerting any attitude control.

Orbital maneuvering

The AOC-0 is not capable of maneuvering in orbit and requires an external service module for all operations, including deorbiting.

Communications

The AOC-0 contains dual Reflectron DP-10 radios. These are enough to maintain communications when in view of the KSC. The system is capable of keeping in contact with the KSC at a distance of 4125 Km and with an HG-5 equipped satellite at 2310 Km.

Antenna ranges

(assumes root model, 0.5 antenna multiplier and 0.5 multiple antenna bonus)

Situation	Range	Unit	Power	Unit	Notes
Single radio	250	Km	0.01	EC/sec	
Dual radios	375	Km	0.02	EC/sec	

To compute effective range, employ the root model equation.

$$R = min(r1, r2) + \sqrt{r1 \times r2}$$

where r1 should be the capsule's range and the r2 range for an HG-5 is 10000Km and KSC is 37500Km.

Science

The capsule has a basic science instrument set containing a barometer and thermometer. These instruments have a storage capacity for one measurement. Data can be moved to the Mk1 pod via EVA or transmitted to KSC via radio.

Transmitting science data imposes a large power drain and should not be done without previous planning.

Re-entry

AOC-0 capsules can re-enter Kerbin’s atmosphere passively. Once contact with the atmosphere is made, the capsule should orient itself in the proper shield first attitude. The shield does not use ablator.

Attaining re-entry attitude requires an external means of propulsion. The AOC-0 capsule is not capable of performing orbital maneuvers.

Landing

Landing is provided by two Mk2 parachutes. The system is redundant and landing can be safely achieved even if a single parachute deploys.

The main parachutes allow a landing speed of 3.4 m/s at sea level. Only one parachute is needed for a safe landing (at about 5.5 m/s). The parachutes open partially at an altitude of 3000 m (0.6 pressure) and fully at 750 metres. They need to be adjusted if the predicted course will cause the landing to happen on mountaineous terrain.

WARNING: Avoid mountains. Plan ahead before executing the re-entry procedure. Consider prematurely opening the parachutes to shorten downrange if necessary.